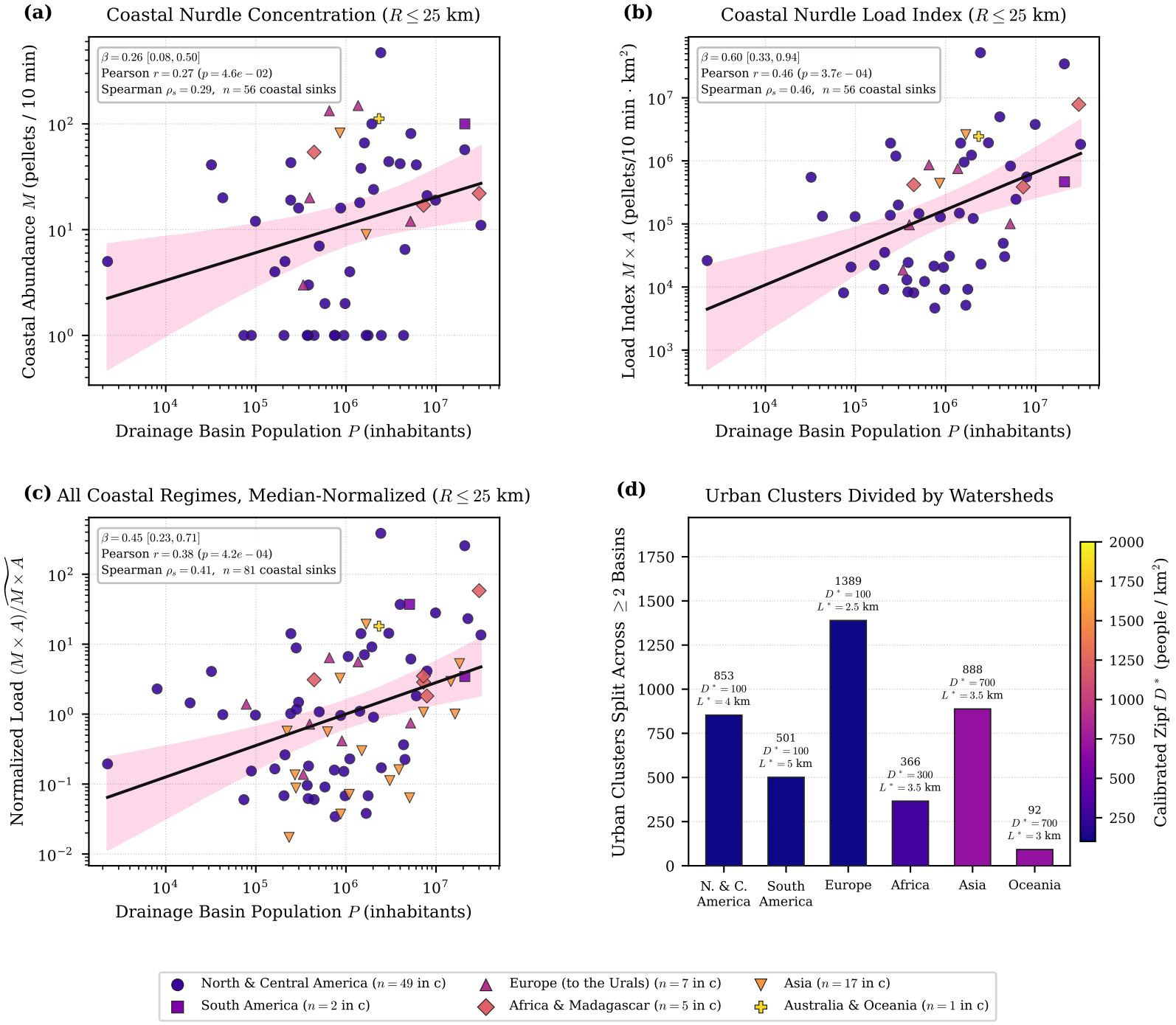

(a)
Coastal Nurdle Concentration ($R \leq 25$ km)
$\beta = 0.26$ [0.08, 0.50]
Pearson $r = 0.27$ ($p = 4.6e-02$)
Spearman $\rho_s = 0.29$, $n = 56$ coastal sinks
Coastal Abundance M (pellets / 10 min)
Drainage Basin Population P (inhabitants)
(b)
Coastal Nurdle Load Index ($R \leq 25$ km)
$\beta = 0.60$ [0.33, 0.94]
Pearson $r = 0.46$ ($p = 3.7e-04$)
Spearman $\rho_s = 0.46$, $n = 56$ coastal sinks
Load Index $M \times A$ (pellets/10 min · km^2)
Drainage Basin Population P (inhabitants)
(c) All Coastal Regimes, Median-Normalized ($R \leq 25$ km)
$\beta = 0.45$ [0.23, 0.71]
Pearson $r = 0.38$ ($p = 4.2e-04$)
Spearman $\rho_s = 0.41$, $n = 81$ coastal sinks
Normalized Load $(M \times A)/\widetilde{M \times A}$
Drainage Basin Population P (inhabitants)
(d)
Urban Clusters Divided by Watersheds
Urban Clusters Split Across ≥ 2 Basins
853
$D^* = 100$
$L^* = 4$ km
501
$D^* = 100$
$L^* = 5$ km
1389
$D^* = 100$
$L^* = 2.5$ km
366
$D^* = 300$
$L^* = 3.5$ km
888
$D^* = 700$
$L^* = 3.5$ km
92
$D^* = 700$
$L^* = 3$ km
N. & C. America
South America
Europe
Africa
Asia
Oceania
Calibrated Zipf D^* (people / km^2)
North & Central America ($n = 49$ in c)
South America ($n = 2$ in c)
Europe (to the Urals) ($n = 7$ in c)
Africa & Madagascar ($n = 5$ in c)
Asia ($n = 17$ in c)
Australia & Oceania ($n = 1$ in c)